

## Incompressible Navier–Stokes

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

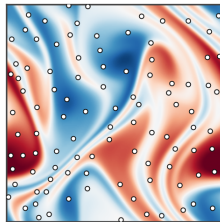
$$\nabla \cdot \mathbf{u} = 0, \quad \mathbf{u} = (u, v)^T$$

$$\mathbf{f} = (A \sin(2\pi k_f y), 0)^T$$

$$\nu = 1/Re, \quad Re = 10^4, \quad k_f = 2$$

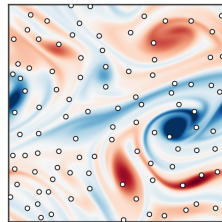
2D Kolmogorov flow,  $256^2$  spectral DNS

## DNS-free sensor placement and measurement



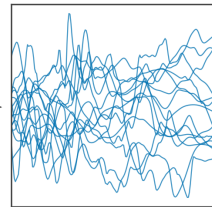
low-cost LES  
QR-pivot on a POD basis

positions only



real flow (DNS stand-in)  
read  $u, v$  at those  $K$

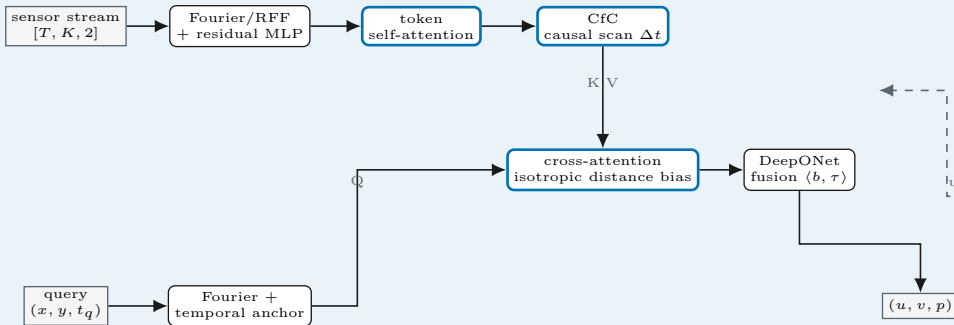
sample



$K = 100$  streams  
the only measured input

## PI-CON operator (this work)

automatic differentiation



## Sensor-only + PDE loss

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_i w_i (\hat{u}_{c_i}(\mathbf{x}_i, t_i; \theta) - u_{c_i}^{\text{obs}})^2$$

$$\mathcal{L}_{\text{NS}, u} = \frac{1}{N_p} \sum_j \mathcal{R}_u^2, \quad \mathcal{L}_{\text{NS}, v} = \frac{1}{N_p} \sum_j \mathcal{R}_v^2$$

$$\mathcal{R}_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \quad \mathcal{L}_{\text{cont}} = \frac{1}{N_p} \sum_j \mathcal{R}_c^2$$

no full-field DNS target enters training

## Weighting + optimiser

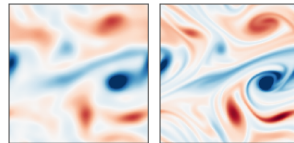
$$\mathcal{L} = \text{GradNorm}(\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{NS}, u}, \mathcal{L}_{\text{NS}, v}, \mathcal{L}_{\text{cont}})$$

$$\lambda \leftarrow \min(\lambda + \beta \mathcal{C}, \Lambda_{\text{max}})$$

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\theta)$$

SOAP + Schedule-Free

## Output and offline check



the DNS full field is used here only to score the result, never to train it